Micronutrients: Vitamins, Minerals, and Common Deficiencies

What Are Micronutrients?

Micronutrients are vitamins and minerals required by the body in small amounts to support a wide range of physiological functions, including immune defense, bone health, nerve signaling, and energy metabolism. Although needed in smaller quantities than macronutrients, deficiencies in micronutrients can lead to significant health problems.

Vitamin D

Vitamin D is unique among vitamins because the body can synthesize it when skin is exposed to sunlight. It plays a central role in calcium absorption and bone health. Deficiency is common in people who spend little time outdoors, live at higher latitudes, or have darker skin pigmentation, which reduces synthesis efficiency.

Symptoms of vitamin D deficiency can include fatigue, bone pain, and muscle weakness. Severe, prolonged deficiency in children can cause rickets, while in adults it can contribute to osteomalacia, a softening of the bones. Good dietary sources include fatty fish, fortified milk, and egg yolks, though supplementation is often recommended in regions with limited sunlight.

Iron

Iron is essential for producing hemoglobin, the protein in red blood cells that carries oxygen throughout the body. Iron deficiency is one of the most common nutritional deficiencies worldwide and can lead to iron-deficiency anemia, characterized by fatigue, pale skin, shortness of breath, and difficulty concentrating.

There are two forms of dietary iron: heme iron, found in animal products and absorbed efficiently by the body, and non-heme iron, found in plant sources such as lentils and spinach, which is absorbed less efficiently. Consuming non-heme iron alongside vitamin C-rich foods, such as citrus fruits, can significantly improve absorption.

Vitamin B12

Vitamin B12 is necessary for red blood cell formation, neurological function, and DNA synthesis. It is found almost exclusively in animal products, which means vegans and strict vegetarians are at elevated risk of deficiency unless they consume fortified foods or take supplements.

B12 deficiency can cause fatigue, weakness, numbness or tingling in the hands and feet, and in severe or prolonged cases, irreversible nerve damage. Because the body stores B12 in the liver, deficiency symptoms may take years to appear after dietary intake drops, making it easy to overlook.

Calcium and Magnesium

Calcium is best known for its role in bone and teeth structure, but it also supports muscle contraction, nerve transmission, and blood clotting. Dairy products, leafy green vegetables, and fortified plant milks are common sources.

Magnesium supports over 300 enzymatic reactions in the body, including those involved in energy production and muscle and nerve function. Nuts, seeds, whole grains, and leafy greens are good dietary sources. Mild magnesium deficiency can present as muscle cramps and fatigue.

Practical Guidance

Most micronutrient needs can be met through a varied diet rich in fruits, vegetables, whole grains, lean proteins, and healthy fats. Individuals with restrictive diets, certain medical conditions, or specific life stages, such as pregnancy, may benefit from targeted supplementation, ideally guided by a blood test and consultation with a healthcare provider rather than guesswork.